Notice, question, predict

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use observations to make a prediction that can be tested.

SUCCESS CRITERIA - I can...

1. Separate supplied facts from statements that have no support.
2. Extend a position pattern and state what your prediction treats as true.
3. Name an observation that could test the prediction.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “evidence” and “prediction” to explain your answer. Sentence starters are optional.

evidence: observations or measurements that support an idea

prediction: what a model says may happen next

model: a drawing, explanation or equation for a physical situation

01 Read a scale

A ruler shows a toy car at **2 cm**, then at **7 cm**.
1. **Draw both positions** and the change.
2. **How far** apart are the marks?

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Delta x = x_f - x_i$; $x_{next} = x_{now} + \Delta x$.
For a repeating position pattern only. i = initial; f = final. **Check the pattern against actual** measurements.

Worked reasoning: A cart is at **2**, **5** and **8 m** at **0**, **1** and **2 s**. Equal jumps of **3 m** suggest **11 m** at **3 s**, if the pattern continues.

Your turn: A motion sensor recorded the first three positions in the table.
1. **Add the predicted position** at **3 s**.
2. **Draw the four positions** on a line and distinguish the prediction from the measurements.

Sentence starter: I predict ... . I used ... to make this prediction because ... .

Time (s)	Position (m)
0	2
1	5
2	8
3	Predict: ____

Marzano check - circle: ① ② ③

03 What does the sensor show?

A class uses a motion sensor to track a cart. The table shows its position once each second. **Describe two patterns** and circle the measurements that support each one.

Support: I notice ____. The data show ____.

Sentence starter: I notice ... . This supports ... because ... .

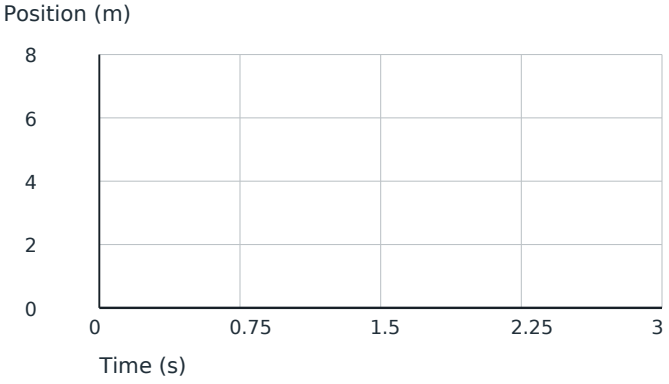
Time (s)	Position (m)
0	1
1	3
2	5

Marzano check - circle: ① ② ③

04 Make a prediction

1. **Use the measurements** in question **3** to **predict where the cart** will be at **3 s**.
2. **Add the prediction** to the blank graph.
3. **Show the pattern you used**.

Sentence starter: I predict ... . I used ... to make this prediction because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 What does the report support?

A fictional space report says: "An object is approaching Earth, but its future path is uncertain." For each statement, write **supported** or **not established**:
(a) It is getting closer.
(b) It will definitely hit Earth.
(c) More observations are needed to **predict its path**.

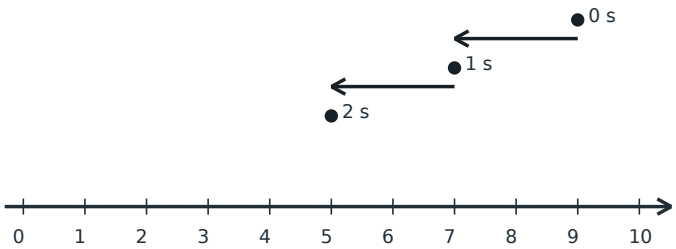
Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

06 Test the prediction

A different cart is observed at **9**, **7**, **5 m** at **0**, **1**, **2 s**.
1. **Predict its position** at **3 s**.
2. What would a new reading of **4 m** make you check?

Sentence starter: Both … . They differ in … because … .



Position (m); right is positive

Marzano check - circle: ① ② ③

07 Ask a question you can investigate

A classmate asks, "Should we worry about the asteroid?" Rewrite this as a question that a measurement could help answer. **Name the measurement** and **explain how it would help**.

Support: I could investigate ____. I would measure ____.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

08 Choose useful measurements

Two groups time the same toy car over the same marked track. One records three times; the other writes "pretty fast." Which record would let someone check the statement? Explain.

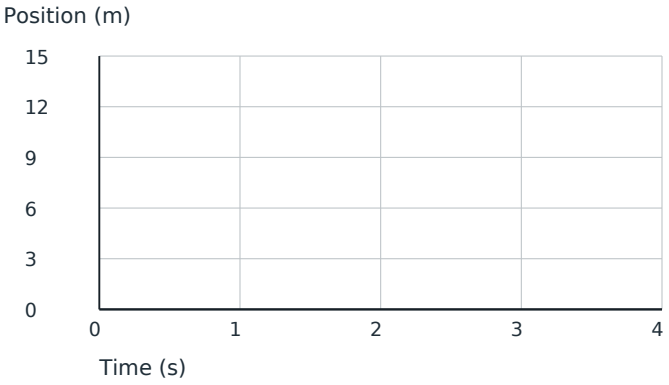
Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

09 Exit check

A robot moves along a marked floor. Its positions are **0**, **3** and **6 m** at **0**, **1** and **2 s**.
(a) Plot the measurements and **predict its position** at **4 s**.
(b) Say what you are treating as true and one later observation that would challenge it.

Sentence starter: I predict … . I used … to make this prediction because … .

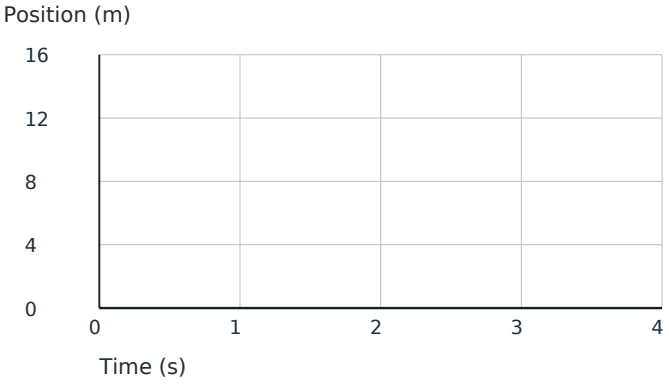


Marzano check - circle: ① ② ③

10 STRETCH: test the prediction

Return to the worked example in problem **2**.
1. Create two possible positions at **4 s** if the cart changes its motion after **2 s**.
2. **Draw both possibilities**.
3. **What measurement** would help choose between them?

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③